

SC Honors Geometry - Nappier
Homework 11.6-11.8 – Pyramids, Cones, and Spheres

Name: _____

Volumes of pyramids.

1. A square pyramid has base side 10 and height 12. Find V.
2. A rectangular pyramid has an 8-by-15 base and height 7. Find V.
3. A triangular pyramid's base has base 6 and height 9; the pyramid is 10 tall. Find V.
4. A pyramid with an 8-by-9 base has volume 240. Find its height.

Cones. Exact and nearest tenth.

5. A cone has radius 8 and height 15. Find the slant height, then the volume.
6. Find the surface area of the same cone.

Spheres.

7. A sphere has radius 7. Find its surface area.
8. Find the volume of the same sphere (nearest tenth).
9. A hemisphere has radius 10. Find its volume (nearest tenth).
10. A medicine capsule is a cylinder of radius 2 and height 6 with a hemisphere on EACH end. Find its total volume (exact and nearest tenth).